

A Spectroscopic Analysis of the Little Red Dots

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1 Introduction and Research Goal

Little Red Dots (LRDs) are a recently identified population of compact, unresolved sources discovered with the *James Webb Space Telescope* (JWST) at high redshift ($z \approx 4\text{--}7$). Their nature remains actively debated: they may be young, dusty star-forming galaxies, obscured active galactic nuclei (AGN), or a combination of both.

LRDs are particularly intriguing due to their distinct V-shaped spectral energy distributions, characterized by a blue ultraviolet (UV) slope and a red optical slope. This may indicate either young stellar populations or dust-reprocessed AGN emission. Additionally, they appear at early cosmic times ($\sim 600\text{ Myr--}1.5\text{ Gyr}$ after the Big Bang), suggesting they may represent a short-lived evolutionary phase, potentially linked to early black hole growth.

The primary goal of this project is to determine whether LRDs host obscured AGN by searching for high-ionization spectral signatures, particularly the [Ne V] $\lambda 3426$ emission line.

2 Data and Sample Selection

2.1 Data Sources

This study uses data from the JWST Advanced Deep Extragalactic Survey (JADES) Data Release 3 in the GOODS-North and GOODS-South fields. The analysis combines:

- NIRCam imaging data and associated photometric catalogs
- NIRSpec spectroscopic data, including both low-resolution prism observations and higher-resolution grating modes (F070LP/G140M, F170LP/G235M, F290LP/G395M)

Photometric fluxes were converted from nJy to AB magnitudes using:

$$m_{\text{AB}} = -2.5 \log_{10}(f_{\text{nJy}}) + 31.4 \quad (1)$$

2.2 Selection Criteria

A three-step selection process was used to identify LRD candidates:

2.2.1 (1) Redshift Selection

Selected sources with $z > 4$ from spectroscopic catalogs.

2.2.2 (2) Color Selection

Following Kocevski et al. (2024), UV and optical slopes (β) were computed:

$$\beta = 0.4 \cdot \frac{m_1 - m_2}{\log_{10}(\lambda_2/\lambda_1)} - 2 \quad (2)$$

Candidates were selected using:

- $\beta_{UV} < -0.37$
- $\beta_{UV} > -2.8$ (removes brown dwarfs)
- $\beta_{opt} > 0$

This isolates the characteristic V-shaped Spectral energy distributions (SEDs) associated with LRDs. Figure 1 shows how the UV and optical slope cuts separate the selected LRD candidates from the full source sample.

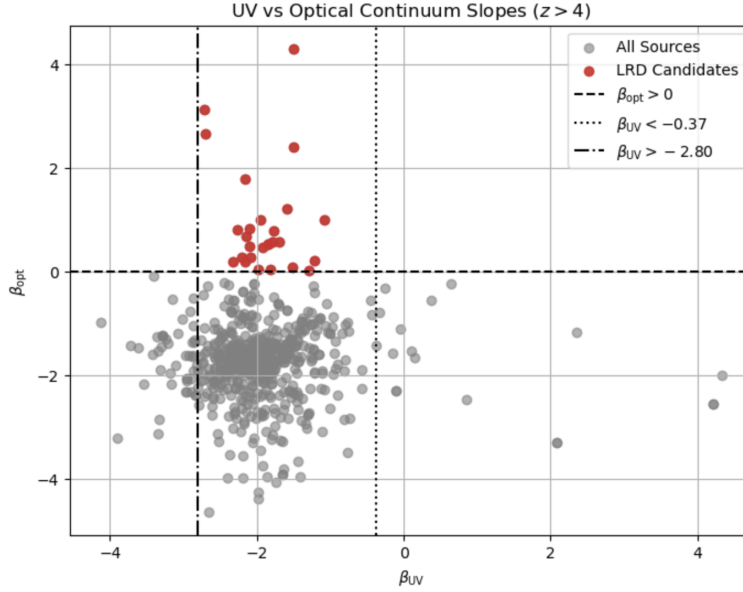


Figure 1: Color-based selection of Little Red Dots (LRDs) using rest-frame UV and optical continuum slopes. Red points indicate selected LRD candidates, while gray points represent all sources. The dashed lines show the selection criteria in β_{UV} and β_{opt} .

2.2.3 (3) Compactness Selection

A stellar locus was constructed using F444W magnitude as a function of half-light radius. The F444W filter was chosen because, at $z \geq 4$, it probes rest-frame optical wavelengths, providing a more reliable tracer of galaxy structure that is less affected by dust attenuation and offers higher signal-to-noise for faint sources.

Sources were classified as point-like if they lay within $1.5\times$ the stellar locus and satisfied a signal-to-noise ratio cut of $SNR > 12$.

Figure 2 illustrates this selection, where LRD candidates are compared to the stellar locus. Sources below the $1.5\times$ threshold are considered compact, consistent with unresolved, point-like objects.

This selection resulted in a total of 24 LRD candidates, of which 22 have available spectra for further analysis.

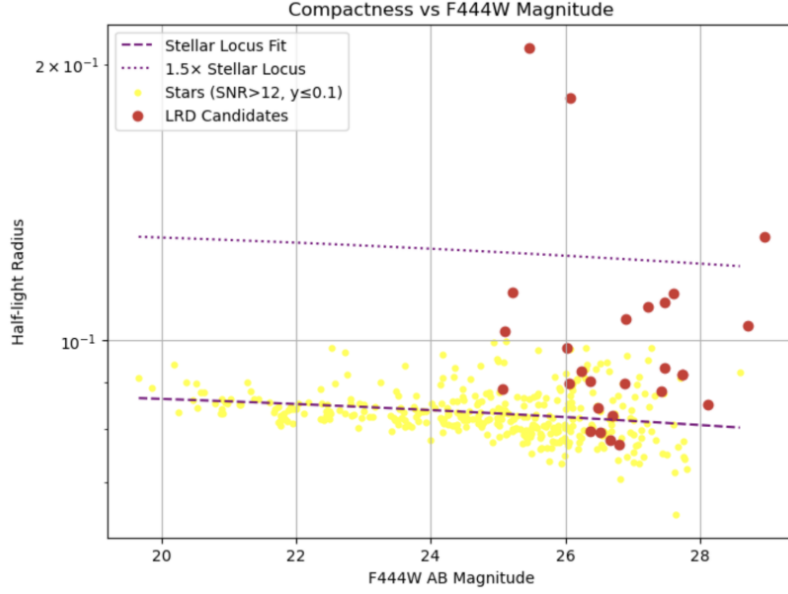


Figure 2: Compactness selection using F444W magnitude and half-light radius. The stellar locus is shown by the dashed line, while the dotted line marks the $1.5\times$ stellar-locus threshold. Yellow points indicate stellar sources used to construct the locus, and red points represent selected LRD candidates. Sources below the dotted line are classified as compact.

3 Spectroscopic Analysis

3.1 Emission Line Fitting

Spectra for the 22 sources were fitted using `PyQSOFit`, targeting key emission lines:

- $H\alpha$, $H\beta$
- O II $\lambda 3727$
- O III $\lambda 5007$
- Ne III $\lambda 3869$
- Ne V $\lambda 3426$
- He II $\lambda 4686$

3.2 [Ne V] as an AGN Diagnostic

The [Ne V] $\lambda 3426$ line is a high-ionization emission feature that requires very high-energy photons and cannot be produced by stellar processes. As a result, it serves as a clean tracer of AGN activity. However, non-detection does not rule out the presence of an AGN, as the line may be too faint to detect given the observational noise.

3.3 Detectability Analysis

To distinguish between intrinsic absence and non-detection due to noise, we extracted the [O III] $\lambda 5007$ flux from spectral fits and used it to estimate the expected [Ne V] flux according to:

$$F_{[\text{Ne V}]} \approx 0.05 \times F_{[\text{O III}]} \quad (3)$$

We then computed the local root-mean-square (RMS) noise near 3426 Å as:

$$\text{RMS} = \sqrt{\langle f^2 \rangle} \quad (4)$$

where f represents the continuum-subtracted flux density in the vicinity of the [Ne V] line. [Ne V] line.

The expected [Ne V] flux was compared to the 1σ , 2σ , and 3σ noise levels. Values exceeding the noise threshold indicate that the line should be detectable, while values within the noise are considered inconclusive.

4 Preliminary Results

The analysis yielded two distinct outcomes. In 15 sources, the expected [Ne V] flux lies within the noise level, making the detection inconclusive. In the remaining 7 sources, the expected flux exceeds the $\sim 1\sigma$ noise level, suggesting that the line should be detectable if present.

These results indicate that a subset of LRDs are strong AGN candidates, while the majority remain ambiguous due to data limitations.

5 Additional Analysis

5.1 Multi-Mode Spectroscopy

To improve constraints on emission line measurements, spectra were examined across different NIRSpec observing modes, including low-resolution prism data and higher-resolution grating modes (G140M, G235M, G395M). The prism provides broad wavelength coverage, while the gratings offer improved spectral resolution over narrower wavelength ranges, allowing for more precise measurements in specific spectral regions.

5.2 Line Ratios and BPT-like Diagram

To further investigate the nature of the sources, we constructed an emission-line diagnostic diagram using ratios of key spectral lines. In particular, we computed $\log([\text{Ne V}]/[\text{Ne III}])$ and $\log([\text{O III}]/\text{H}\beta)$ using integrated fluxes with propagated uncertainties.

Figure 3 shows the resulting BPT-like diagram. Sources derived from prism observations and grating data are shown separately, enabling comparison between different spectral resolutions. Triangular markers indicate upper limits for sources with non-detections, with the direction of the marker representing the constraint. The dashed lines indicate approximate boundaries between star-forming (SF), composite, and AGN regions.

The distribution of sources suggests that several occupy regions consistent with AGN or composite activity, supporting the interpretation that at least a fraction of LRDs host active nuclei. However, uncertainties remain significant for many sources, particularly those with low signal-to-noise ratios.

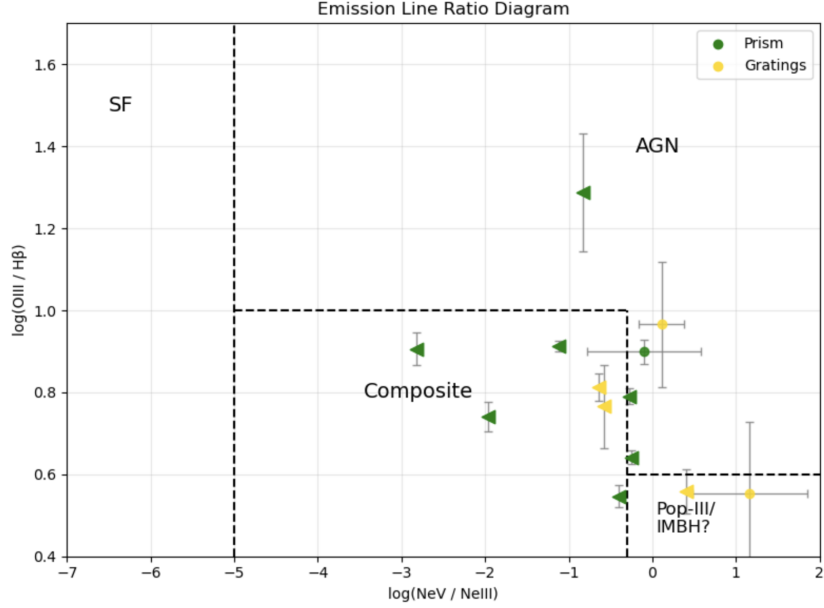


Figure 3: BPT-like diagnostic diagram using $\log([Ne V]/[Ne III])$ and $\log([O III]/H\beta)$. Green points correspond to prism observations, while yellow points represent grating data. Triangles denote upper limits for sources with non-detections, with the direction indicating the constraint. The dashed lines indicate approximate boundaries between star-forming (SF), composite, and AGN regions. A region associated with extreme ionization conditions, potentially consistent with Population III stars or intermediate-mass black holes (IMBH), is indicated.

6 Interpretation

The results suggest that LRDs are not a homogeneous population. Some sources are consistent with hosting obscured AGN, while others may be purely star-forming or too faint to classify given current detection limits. The presence of sources where $[Ne V]$ should be detectable supports AGN activity in at least a fraction of the population. However, the large number of inconclusive cases highlights the importance of signal-to-noise limitations in interpreting high-redshift spectra.

7 Future Work

Future work will focus on improving the BPT-like diagram by refining the MCMC fitting procedure to obtain more reliable upper limits. In addition, stacking spectra will be implemented to boost the signal-to-noise ratio (SNR), allowing faint emission lines to become detectable and enabling a more robust assessment of whether AGN signatures are present but buried.

Expanding the sample using additional surveys, such as CEERS, will further improve statistical significance and help determine whether the observed trends are representative of the broader LRD population.